Lecture 23

Convolution

Consider the IVP

$$y'' + y = g(t), \quad y(0) = 0, \quad y'(0) = 0$$

If we wish to solve this equation using Laplace transforms we end up with the following expression for $Y(s)$:

$\mathcal{L}\{g(t)\}$

$$\mathcal{L}\{y''+y=g(t)\} \Rightarrow (s^2Y(s) - s\cancel{y(0)}^{0} - \cancel{y'(0)}^{0}) + Y(s) = G(s)$$

$$\Rightarrow s^2Y(s) + Y(s) = G(s)$$

$$\Rightarrow Y(s)(s^2+1) = G(s) \Rightarrow Y(s) = \frac{G(s)}{s^2+1}$$

char eqn = 0

$$\Rightarrow Y(s) = G(s) \cdot \frac{1}{s^2+1}$$

We see that in order to find $y(t)$ we will need to determine the inverse Laplace transform of the product

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}G(s)\right\}.$$

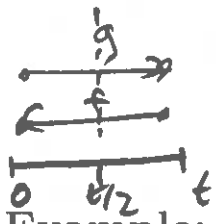
Is there a way we can find the inverse transform of a product? Note that in general

$$\mathcal{L}^{-1}\{F(s)G(s)\} \neq \mathcal{L}^{-1}\{F(s)\}\,\mathcal{L}^{-1}\{G(s)\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\cdot\frac{1}{s}\right\} \neq \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}\cdot\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t \neq 1 \cdot 1 = 1$$

Definition: Let $f(t)$ and $g(t)$ be piecewise continuous on $[0, \infty)$. The **convolution** of $f(t)$ and $g(t)$ is defined as



$$(f * g)(t) = \int_0^t f(t-v)g(v)\, dv$$

not $0 \to \infty$!!

convolution

Note: This is not a Laplace transform

Example: Evaluate $(t * 1)(t)$.

$$(t * 1)(t) = \int_0^t \underbrace{(t-v)}_{f(t-v)} \cdot 1\, dv = \int_0^t (t-v)\, dv$$

f g

$$= \left[tv - \tfrac{1}{2}v^2\right]\Big|_{v=0}^{v=t}$$

$$= \left[t(t) - \tfrac{1}{2}(t)^2\right] - \left[t(0) - \tfrac{1}{2}0^2\right] = t^2 - \tfrac{1}{2}t^2 = \tfrac{1}{2}t^2$$

We can see that the convolution of two functions is different from their multiplication as

$$t \cdot 1 \neq (t * 1)(t) = \frac{t^2}{2}$$

Theorem (Properties of Convolution)**:** Let $f(t)$, $g(t)$, and $h(t)$ be piecewise continuous on $[0, \infty)$. Then

1. $f * g = g * f$ (commutativity)
2. $f * (g + h) = (f * g) + (f * h)$ (distributivity)
3. $(f * g) * h = f * (g * h)$ (associativity)
4. $f * 0 = 0$ (null element) zero

$$\int_0^t f(t-v) \cdot 0\, dv = \int_0^t 0\, dv = 0$$

multiplication
$f \cdot g = g \cdot f$
$f \cdot 0 = 0$
$f \cdot 1 = f$
addition
$f + g = g + f$
$f + 0 = f$
$f - f = 0$

identity elements

Theorem (Convolution Theorem): Let $f(t)$ and $g(t)$ be piecewise continuous on $[0,\infty)$ and of exponential order α. If $\mathscr{L}\{f(t)\} = F(s)$ and $\mathscr{L}\{g(t)\} = G(s)$, then

$$\mathscr{L}\{f * g\} = F(s)G(s) \iff \underbrace{\mathscr{L}^{-1}\{F(s)G(s)\} = f*g}_{!!}$$

Proof:

$$\mathscr{L}\{f*g\} = \int_0^\infty \left[\int_0^t f(t-v)g(v)\,dv\right] e^{-st}\,dt$$

$$u(t-v) = \begin{cases} 0, & t-v<0 \\ 1, & t-v>0 \end{cases} = \begin{cases} 0, & t<v \\ 1, & t>v \end{cases}$$

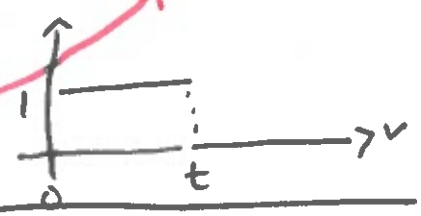
$$= \int_0^\infty \left[e^{-st}\int_0^t f(t-v)\,g(v)\,dv\right] dt$$

$$= \int_0^\infty \left[e^{-st}\int_0^{\cancel{t}\,\infty} f(t-v)\cdot u(t-v)\,g(v)\,dv\right] dt$$

reverse

$0 \le v \le t$

$v < t$



$$= \int_0^\infty g(v)\left[\int_0^\infty e^{-st} f(t-v)\,u(t-v)\,dt\right] dv$$

use $w = t - v \Rightarrow t = w + v$

$dw = dt$

$$= \int_0^\infty g(v)\left[\int_0^\infty e^{-s(w+v)} f(w)\,u(w)\,dw\right] dv$$

$$u(w) = \begin{cases} 0, & w<0 \\ 1, & w>0 \end{cases}$$

$$= \int_0^\infty g(v)\left[\int_0^\infty \underbrace{e^{-sv}}_{\text{constant}} e^{-sw} f(w)\,u(w)\,dw\right] dv$$

$$= \int_0^\infty g(v)e^{-sv}\left[\underbrace{\int_0^\infty e^{-sw} f(w)\cdot 1\,dw}_{F(s)}\right] dv$$

$$= \int_0^\infty g(v)e^{-sv}F(s)\,dv = F(s)\underbrace{\int_0^\infty e^{-sv} g(v)\,dv}_{G(s)} = F(s)G(s) \quad \square$$

$$\boxed{\mathscr{L}^{-1}\{F(s)G(s)\} = \mathscr{L}^{-1}\{F(s)\} * \mathscr{L}^{-1}\{G(s)\}}$$

Example: Solve $y'' + y = g(t)$, $y(0) = 0$, $y'(0) = 1$.

$$\mathcal{L}\{y''+y = g(t)\} \Rightarrow s^2Y(s) - s\cancel{y(0)}^{0} - \cancel{y'(0)}^{1} = G(s)$$

$$\Rightarrow (s^2+1)Y(s) - 1 = G(s) \Rightarrow Y(s) = \frac{G(s)+1}{s^2+1}$$

$$\Rightarrow Y(s) = G(s)\cdot\frac{1}{s^2+1} + \frac{1}{s^2+1}$$

$$\Rightarrow y(t) = g(t) * \sin(t) + \sin(t) = \boxed{\underbrace{\sin(t)}_{\text{homogeneous}} + \underbrace{\int_0^t g(t-v)\sin(v)\,dv}_{\text{particular}}}$$

Example: Evaluate $\mathcal{L}\left\{\int_0^t e^v \sin(t-v)\,dv\right\}$

$$f*g = \int_0^t f(t-v)g(v)\,dv$$

$$\mathcal{L}\left\{\int_0^t \sin(t-v)e^v\,dv\right\} = \mathcal{L}\{\sin(t) * e^t\}$$

$$= \mathcal{L}\{\sin(t)\}\cdot\mathcal{L}\{e^t\}$$

$$= \frac{1}{s^2+1}\cdot\frac{1}{s-1} \quad \left(= \frac{1}{(s-1)(s^2+1)}\right)$$

$$\mathcal{L}\{\sin(t)\} = \frac{1}{s^2+1}$$

Example: Evaluate $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\cdot\frac{1}{s^2+1}\right\}$

$$= \mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} * \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \cos(t) * \sin(t)$$

$$= \int_0^t \cos(\underbrace{t-v}_{b})\sin(\underbrace{v}_{a})\,dv$$

identity

$$\boxed{\begin{aligned}&\sin(a)\cos(b) = \tfrac{1}{2}[\sin(a+b) + \sin(a-b)]\\ &\text{or}\\ &\cos(A-B) = \cos(A)\cos(B) + \sin(A)\sin(B)\end{aligned}}$$

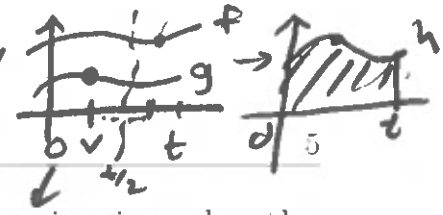
$$= \int_0^t \tfrac{1}{2}[\sin(v+(t-v)) + \sin(v-(t-v))]\,dv$$

$$= \int_0^t \tfrac{1}{2}[\sin(t) + \sin(2v-t)]\,dv = \left[\tfrac{1}{2}\sin(t)\cdot v - \tfrac{1}{2}\tfrac{1}{2}\cos(2v-t)\right]\Big|_{v=0}^{v=t}$$

$$= \left[\tfrac{1}{2}\sin(t)\cdot t - \tfrac{1}{4}\cancel{\cos(t)}\right] - \left[\tfrac{1}{2}\cancel{\sin(t)}^{0}\cdot 0 - \tfrac{1}{4}\cancel{\cos(-t)}\right] = \tfrac{1}{2}t\sin(t)$$

$\cos(t)$

$$f*g = \int_0^t \underbrace{f(t-v)g(v)}_{h(v)}\,dv = \int_0^t h(v)\,dv$$



Recall that the **accumulation function** of $f(t)$ at $t = a$ is given by the integral

$$\int_a^t f(v)\,dv$$

$g(x) = \int_a^x f(t)\,dt$

$g'(x) = f(x)$

By the Fundamental Theorem of Calculus, the accumulation function of $f(t)$ is an antiderivative of $f(t)$. We can see that

$$(1 * f)(t) = \int_0^t 1 \cdot f(v)\,dv = \int_0^t f(v)\,dv$$

and so

$$\mathscr{L}\left\{\int_0^t f(v)\,dv\right\} = \mathscr{L}\{1 * f\} = \mathscr{L}\{1\} \cdot \mathscr{L}\{f(t)\} = \frac{1}{s} \cdot F(s) = \frac{F(s)}{s}$$

Example: Evaluate $\mathscr{L}^{-1}\left\{\dfrac{1}{s(s^2+1)}\right\}$.

Recall $\mathscr{L}^{-1}\{\ln(s-a)\}$

anti of $\frac{1}{s-a}$

$\frac{e^{at}}{t}$

$= \mathscr{L}^{-1}\left\{\frac{1}{s} \cdot \frac{1}{s^2+1}\right\}$ $f(t) = \sin(t)$

$= \mathscr{L}^{-1}\left\{\frac{1}{s} \cdot F(s)\right\}$

$= 1 * \sin(t) = \int_0^t \sin(v)\,dv$

$= -\cos(v)\Big|_{v=0}^{v=t} = -\cos(t) - (-\cos(0))$

$= 1 - \cos(t)$

$(-\cos(x) + C)$

By including the accumulation function of $y(t)$ in a differential equation for y, we can solve equations with both derivatives and antiderivatives of some unknown function y. These are called **integro-differential equations**. We can also include regular convolutions of two functions as well.

Example: Solve $y(t) = 4t + \underbrace{\int_0^t y(t-v)\sin(v)\,dv}_{y(t)*\sin(t)}$

$t=0 \Rightarrow y(0) = 4\cdot 0 + \int_0^0 y(0-v)\sin(v)\,dv$

$\Rightarrow y(0)=0$

$\Rightarrow Y(s) = \frac{4}{s^2} + Y(s)\cdot\frac{1}{s^2+1}$

$\Rightarrow Y(s) - Y(s)\frac{1}{s^2+1} = \frac{4}{s^2} \Rightarrow \left[1-\frac{1}{s^2+1}\right]Y(s) = \frac{4}{s^2}$

$\frac{s^2+1-1}{s^2+1} = \frac{s^2}{s^2+1}$

$\Rightarrow Y(s) = \frac{4}{s^2}\cdot\frac{s^2+1}{s^2} = \frac{4s^2+4}{s^4}$

$\Rightarrow Y(s) = \frac{4s^2}{s^4} + \frac{4}{s^4} = \frac{4}{s^2} + \frac{4}{s^4} \Rightarrow y(t) = 4t + 4\cdot\frac{1}{6}t^3 = 4t + \frac{2}{3}t^3$

$\mathcal{L}\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}$

$\Rightarrow \boxed{y(t) = 4t + \frac{2}{3}t^3}$

Example: Solve $y' + y + \int_0^t y(v)\,dv = 1, \quad y(0) = 0$

$\boxed{\mathcal{L}\{\int_0^t y(v)dv\} = \frac{Y(s)}{s}}$

$\xrightarrow{t=0} y'(0) + \cancel{y(0)}^{\to 0} + \int_0^0 y(v)dv = 1$

$\Rightarrow y'(0) = 1$

$\Rightarrow [sY(s) - \cancel{y(0)}^{\to 0}] + Y(s) + \frac{Y(s)}{s} = \frac{1}{s}$

$\Rightarrow Y(s)\left(s + 1 + \frac{1}{s}\right) = \frac{1}{s} \Rightarrow Y(s)\left[\frac{s^2+s+1}{s}\right] = \frac{1}{s}$

$\Rightarrow Y(s) = \frac{s}{s(s^2+s+1)} = \frac{1}{s^2+s+1}$

$\Rightarrow Y(s) = \frac{1}{(s^2+s+\frac{1}{4})-\frac{1}{4}+1} = \frac{1}{(s+\frac{1}{2})^2+\frac{3}{4}} = \frac{1}{\frac{\sqrt{3}}{2}}\cdot\frac{\sqrt{3}/2}{(s+\frac{1}{2})^2+(\frac{\sqrt{3}}{2})^2}$

$a = -\frac{1}{2} \quad b = \frac{\sqrt{3}}{2}$

$\Rightarrow \boxed{y(t) = \frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)}$

$y' + P(x)y = Q(x)$

$\Rightarrow y = \frac{1}{\mu(x)}\int \mu(x)Q(x)\,dx, \quad \mu(x) = e^{\int P(x)dx}$

When we studied first-order linear equations we wrote down a general formula for a solution regardless of the functions involved. Using convolutions, we can now do the same for linear constant coefficient equations.

We will consider the theory for the second-order case, but the general nth-order case follows in the exact same manner.

Suppose we wish to solve some equation

$$ay'' + by' + cy = g(t), \quad y(0) = 0, \quad y'(0) = 0. \tag{23.1}$$

If we take the Laplace transform of both sides we have

$$\begin{aligned}\mathcal{L}\{ay'' + by' + cy\} = \mathcal{L}\{g(t)\} &\Rightarrow as^2Y(s) + bsY(s) + cY(s) = G(s) \\ &\Rightarrow (as^2 + bs + c)Y(s) = G(s) \\ &\Rightarrow Y(s) = \frac{G(s)}{as^2 + bs + c} = G(s)\cdot\frac{1}{as^2+bs+c}\end{aligned}$$

$\cancel{-sy(0) - y'(0)}$ 0 $\cancel{-y(0)}$ 0

char eqn $=0$

Definition: If $\mathcal{L}\{y(t)\} = Y(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, we define the **transfer function**, $H(s)$, as

$$H(s) = \frac{Y(s)}{G(s)} = \boxed{\frac{1}{as^2 + bs + c}} \quad \frac{\cancel{G(s)}}{\cancel{G(s)}}$$

We then say that $\mathcal{L}^{-1}\{H(s)\} = h(t)$ is the **impulse response function**.

Note: The transfer function is just the reciprocal of the characteristic polynomial of the associated homogeneous equation.

Therefore, for $ay'' + by' + cy = g(t)$, $y(0) = 0$, $y'(0) = 0$ we have that

$$Y(s) = \frac{G(s)}{as^2 + bs + c} = H(s)G(s)$$

$G(s)\cdot\frac{1}{as^2+bs+c}$

$G(s)H(s)$

and so

$$y(t) = \mathscr{L}^{-1}\{Y(s)\} = \mathscr{L}^{-1}\{H(s)G(s)\} = h * g$$

forcing function

impulse response

is the solution to (23.1) when $y(0) = 0$ and $y'(0) = 0$.

$ay'' + by' + cy = g(t)$

When we have nonzero initial conditions, then our homogeneous solution is nonzero. It follows that if we wish to solve the IVP

$$ay'' + by' + cy = g(t), \quad y(0) = y_0, \quad y'(0) = y_1,$$

if $y_h(t)$ is the solution to the associated homogeneous equation (with the same initial conditions), then

$ay'' + by' + cy = 0$ ( )

$$y_g(t) = y_h(t) + (h * g)(t) \rightarrow \int_0^t$$

is the general solution to the equation where $h(t)$ is its associated impulse response function.

$y(0) = 0$
$y'(0) = 0$! !

Note that:

$$\left.\begin{aligned} y(0) &= y_h(0) + (h * g)(0) = y_0 + 0 = y_0 \\ y'(0) &= y_h'(0) + (h * g)'(0) = y_1 + 0 = y_1 \end{aligned}\right\}$$

Example: Solve $y'' + 2y' + 5y = g(t)$, $y(0) = 2$, $y'(0) = 2$

$ay'' + by' + cy = g(t)$, $y(0) = y_0$, $y'(0) = y_1$ $\Rightarrow y(t) = y_h(t) + (h * g)(t)$

$a = 1$, $b = 2$, $c = 5$, $y_0 = 2$, $y_1 = 2$

(1) Solve the homogeneous eqn.

(2) Find $h(t)$ impulse response function

(3) Setup y_g.

char eqn: $r^2 + 2r + 5 = 0$ $\quad r = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm 4i}{2}$

$(r+1)^2 + 4 = 0$ $\quad r = -1 \pm 2i$

$y_h = c_1 e^{-t}\cos(2t) + c_2 e^{-t}\sin(2t)$

Instead $\mathcal{L}\{y''+2y'+5y=0\} \Rightarrow [s^2Y(s)-sy(0)-y'(0)] + 2[sY(s)-y(0)] + 5Y(s) = 0$

$$\Rightarrow s^2Y(s)-2s-2+2sY(s)-4+5Y(s)=0$$

$$\Rightarrow [s^2+2s+5]Y(s)-2s-6=0$$

$$\Rightarrow Y(s)=\frac{2s+6}{s^2+2s+5}=\frac{2s+6}{(s+1)^2+4}=\frac{2(s+1)+4}{(s+1)^2+4}$$

$2s+2+4$

$$\Rightarrow Y(s)=2\cdot\frac{s+1}{(s+1)^2+4}+2\cdot\frac{2}{(s+1)^2+4} \Rightarrow y_h(t)=2e^{-t}\cos(2t)+2e^{-t}\sin(2t)$$

$$\Rightarrow \boxed{y_h(t)=2e^{-t}\cos(2t)+2e^{-t}\sin(2t)}$$

Next, $h(t)=\mathcal{L}^{-1}\{H(s)\}=\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s+5}\right\}$

$$=\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+4}\right\}=\mathcal{L}^{-1}\left\{\frac{1}{2}\cdot\frac{2}{(s+1)^2+4}\right\}$$

$$\Rightarrow \boxed{h(t)=\frac{1}{2}e^{-t}\sin(2t)}$$

$(\frac{1}{2}e^{-t}\sin(2t))*(g(t))$

So $y_g=y_h(t)+(h*g)(t)$

$$\Rightarrow \boxed{y_g=\left(2e^{-t}\cos(2t)+2e^{-t}\sin(2t)\right)+\int_0^t \frac{1}{2}e^{-(t-v)}\sin(2(t-v))g(v)\,dv}$$